Capstone Data Exploration

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To do:

- Make sure its same type of measurement as a2016 data (ex real/nominal, ppp, etc)
- get steel data
- clean
- join
- get concrete/cement data
- clean
- join
- data explore
  - **remove outlier (usa), redo var relations explore**

Setup

```
# vector of needed packages
pkgs <- c("tidyverse", "janitor", "countrycode", "readxl")
# install packages that are on needed list but not on list of installed
install.packages(setdiff(pkgs, rownames(installed.packages())))

library(tidyverse)
library(janitor) # to clean column names via clean_names()
library(countrycode) # to unify country codes to one format
library(readxl) # to import one sheet of multi-sheet excel file
```

```r
## Source: NYU Marron Institute of Urban Management. (2025b, January 27). "Data".
##         Transit Costs Project. https://transitcosts.com/data/
transit_data <- read.csv("data/transit_costs_project_data.csv") %>%
  clean_names() %>% # standardizes column name format (lowercase, _ as seperator)
  mutate(across(where(is.character), \(x) na_if(x, ""))) # "" values -> NA

## Source: University of Ottawa. (2008). Alphabetical Index of the Political
##         Entities and Corresponding Legal Systems . Juriglobe.
##         https://www.juriglobe.ca/eng/sys-juri/index-alpha.php
legal_origins_data <- read.csv("data/legal_origins_data.csv") %>%
  clean_names() %>%
  mutate(across(where(is.character), \(x) na_if(x, "")))

## Source: Hensel, P. R. (2018, November 13). ICOW Colonial History Data.
##         https://paulhensel.org/icowcol.html
colonize_data <- read.csv("data/col_data.csv") %>%
  clean_names() %>%
  mutate(across(where(is.character), \(x) na_if(x, "")))

## Source: World Bank Group. (2026, October 8). World development indicators.
##         DataBank.
##         https://databank.worldbank.org/source/world-development-indicators
wb_data <- read.csv("data/wb_gdppp_gov_data.csv") %>%
  clean_names() %>%
  mutate(across(where(is.character), \(x) na_if(x, "")))

## Source: International Labour Orginization. (n.d.). Labour statistics for ILO
##         sectors. ILOSTAT. ilostat.ilo.org/databases/ilo-sectors/
wage_data <- read.csv("data/const_wage_data.csv") %>%
  clean_names() %>%
  mutate(across(where(is.character), \(x) na_if(x, "")))

## Source: World Bank Group. (2025, April 24). Global fuel prices database - world
##         bank data catalog. Data Catalog.
##         https://datacatalog.worldbank.org/search/dataset/0066829/global-fuel-prices-datab
diesel_a2016_data <- read_excel("data/Global_Fuel_Prices_Database.xlsx",
                          sheet = "Diesel USD") %>%
  rename_with(\(nm) as.character(excel_numeric_to_date(as.numeric(nm))),
              .cols = matches("^\\d+$")) %>%
  clean_names() %>%
  mutate(across(where(is.character), \(x) na_if(x, "")))
```

```
## Source: World Bank Group. (n.d.). WDI Database Archives. DataBank.
##         https://databank.worldbank.org/source/wdi-database-archives#
diesel_b2016_data <- read.csv("data/wb_diesel_b2016_data.csv") %>%
  clean_names() %>%
  mutate(across(where(is.character), \(x) na_if(x, "")))
```

Data Cleaning

Merger preparation

```
# From countrycode: detects country code type
### guess_field(transit_data$country) # iso2c: 97% match

# Add variable that contains the country's iso3c code
transit_data_1 <- transit_data %>%
  mutate(iso3c = countrycode(country, "iso2c", "iso3c",
                             custom_match = c("UK" = "GBR", "DR" = "DOM"))) %>%
  relocate(iso3c, .after = country)
```

```
# default min = 80%
# guess_field(toupper(legal_origins_data$political_entities), min_similarity = 0)
# no match, use "country.name"

# Claude (Opus 5.5) code
legal_origins_data_1 <- legal_origins_data %>%
  mutate(
    name_clean = gsub("\\s*\\(minus[^)]*\\)", "", political_entities),
    name_clean = gsub("\\s+", " ", name_clean),
    iso3c = countrycode(name_clean, "country.name", "iso3c",
                        custom_match = c("GUAM (USA)" = "GUM",
                                         "MARIANA (USA)" = "MNP",
                                         "SAMOA, AMERICAN (USA)" = "ASM",
                                         "VIRGIN ISLANDS (USA)" = "VIR",
                                         "PORTO RICO (ASS. USA)" = "PRI",
                                         "MICRONESIA" = "FSM",
                                         "HAWAII (USA)" = NA,
                                         "LOUISIANA (USA)" = NA,
                                         "NORTHERN IRELAND (UK)" = NA))) %>%
  relocate(iso3c, .after = political_entities)
```

```r
# From countrycode: detects country code type
# guess_field(colonize_data$state) # cown: 99% match


# Austro-hungarian Empire is the sole colonizer w/out an iso3c code
# 3 tansit projects are in czechia, colonized by the austro-hungarian empire
# Using Austria iso3c code: shares a boarder with Czechia (Hungary does not)

# Add variable that contains the country's iso3c code
# Add variable that contains the colonizor's iso3c code
# Add variable that contains the ind_from's iso3c code
colonize_data_1 <- colonize_data %>%
  mutate(iso3c = countrycode(state, "cown", "iso3c")) %>%
  relocate(iso3c, .after = state) %>%
  mutate(col_ruler_iso3c = countrycode(col_ruler, "cown", "iso3c",
                                       custom_match = c("300" = "AUT"))) %>%
  relocate(col_ruler_iso3c, .after = col_ruler) %>%
  mutate(ind_from_iso3c = countrycode(ind_from, "cown", "iso3c")) %>%
  relocate(ind_from_iso3c, .after = ind_from)
```

```r
# Rename year headers to just 4 digit year
wb_data_1 <- wb_data %>%
  rename_with(\(x) substr(x, 2, 5), .cols = starts_with("x"))

wb_data_1 <- wb_data_1 %>%
  filter(!is.na(country_code)) %>%
  filter(!is.na(series_name))

# Lengthen df by year
wb_data_1 <- wb_data_1 %>%
  pivot_longer(cols = '1960':'2025',
               names_to = "year",
               values_to = "value")

# Remove all character in series_name values from '-' on
wb_data_1 <- wb_data_1 %>%
  mutate(series_name = trimws(gsub("-.*", "", series_name)))

# Remove series_code column (to widen more easily)
wb_data_1 <- wb_data_1 %>%
  select(-'series_code')
```

```r
# Widen df by series
wb_data_1 <- wb_data_1 %>%
  pivot_wider(names_from = 'series_name', values_from = 'value')

# apply clean names to new headers from pivot_wider (series_code values)
wb_data_1 <- wb_data_1 %>%
  clean_names()

# guess_field(wb_data_1$country_code)
# wb code nearly but not fully identical to iso3c

# Add variable that contains the country's iso3c code
wb_data_1 <- wb_data_1 %>%
  mutate(iso3c = countrycode(country_code, "wb", "iso3c",
                             custom_match = c(XKX = "XKX", CHI = NA))) %>%
  relocate(iso3c, .after = country_code)

# Change 'year' class, chr -> int (to match transit data in left_join)
wb_data_1 <- wb_data_1 %>%
  mutate(year = as.integer(year))
```

```r
# filter to relevent observations
wage_data_1 <- wage_data %>%
  filter(sex_label == "Total") %>%
  filter(classif1_label == "ILO sector: Construction") %>%
  filter(classif2_label == "Currency: 2021 PPP $")

# check # of unique nation-year combonations
# wage_data_1 %>%
#   distinct(ref_area_label, time) %>%
#   count()
    # Equal to obs count in wage_data_1: there is now only 1 obs per
    # nation-year combanation

# Check # lost countries after filtering
# unique(wage_data$ref_area_label) %>%
#   length()
#
# unique(wage_data_1$ref_area_label) %>%
#   length()
# -7
```

```r
# guess_field(wage_data_1$ref_area_label)
# 92.3% match for   un.name.en

wage_data_1 <- wage_data_1 %>%
  mutate(iso3c = countrycode(ref_area_label, "country.name", "iso3c")) %>%
  relocate(iso3c, .after = ref_area_label) %>%
  rename(year = time) %>%
  rename(wage_per_month = obs_value)
```

```r
# There are multiple observations per country, default_map_1_show seems to indicate
# the main observation

# filter to only default_map_1_show == 1
diesel_a2016_data_1 <- diesel_a2016_data %>%
  filter(default_map_1_to_show == 1)



# create average value for years
start <- 2016
n_times <- 10
for (yr in start:(start + n_times - 1)) {
  diesel_a2016_data_1 <- diesel_a2016_data_1 %>%
    mutate("{yr}" := rowMeans(pick(contains(as.character(yr))), na.rm = TRUE))
}

# delete monthly values
diesel_a2016_data_1 <- diesel_a2016_data_1 %>%
  select(-c("x2015_12_01":"x2025_04_01"))

# Lengthen df by year
diesel_a2016_data_1 <- diesel_a2016_data_1 %>%
  pivot_longer(cols = '2016':'2025',
               names_to = "year",
               values_to = "diesel_usd_per_liter_a")

# guess_field(diesel_a2016_data_1$country_code)
# diesel_a2016_data_1 <- diesel_a2016_data_1 %>%
#   mutate(iso3c = countrycode(country_code, "wb", "iso3c"))
# Only Curacao is not iso3c already

# correct curacao code and rename country_code to iso3c
diesel_a2016_data_1 <- diesel_a2016_data_1 %>%
```

```r
  mutate(country_code = if_else(country_code == "ANT", "CUW", country_code)) %>%
  rename(iso3c = country_code)

# make year an int variable
diesel_a2016_data_1 <- diesel_a2016_data_1 %>%
  mutate(year = as.integer(year))

# fix french guiana, martinique, and moldova iso3c codes
diesel_a2016_data_1 <- diesel_a2016_data_1 %>%
  mutate(iso3c = if_else(diesel_lcu_liter == "French Guiana", "GUF", iso3c)) %>%
  mutate(iso3c = case_when(country == "Martinique" ~ "MTQ",
                           country == "Moldova" ~ "MDA",
                           .default = iso3c))
```

```r
# regex by claude
# renames messy year headers to just the four digit year
diesel_b2016_data_1 <- diesel_b2016_data %>%
  rename_with(\(nm) sub("^x(\\d{4})_yr\\d{4}$", "\\1", nm),
              .cols = matches("^x\\d{4}_yr\\d{4}$"))

#CHOICE#
# pivot_longer by year, fills in missing year values by averaging nearest 1 or 2 value(s) of
diesel_b2016_data_1 <- diesel_b2016_data_1 %>%
  pivot_longer(`1960`:`2015`, names_to = "year", values_to = "diesel_usd_per_liter_b") %>%
  mutate(diesel_usd_per_liter_b = as.numeric(diesel_usd_per_liter_b),
         year = as.numeric(year)) %>%
  mutate(prev = diesel_usd_per_liter_b, nxt = diesel_usd_per_liter_b,
         prev_yr = if_else(is.na(diesel_usd_per_liter_b), NA, year), nxt_yr = prev_yr) %>%
  group_by(country_name) %>%
  fill(prev, prev_yr, .direction = "down") %>%
  fill(nxt, nxt_yr, .direction = "up") %>%
  ungroup() %>%
  mutate(prev = if_else(year - prev_yr <= 2, prev, NA),
         nxt = if_else(nxt_yr - year <= 2, nxt, NA),
         diesel_usd_per_liter_b = coalesce(diesel_usd_per_liter_b, (prev + nxt) / 2, prev, nx
  select(-prev, -nxt, -prev_yr, -nxt_yr)
```

```
Warning: There was 1 warning in `mutate()`.
i In argument: `diesel_usd_per_liter_b = as.numeric(diesel_usd_per_liter_b)`.
Caused by warning:
! NAs introduced by coercion
```

```r
# guess_field(diesel_b2016_data_1$country_code, min_similarity = 50)
# ~75% match for iso3c, genc3cm, wb, wb_api3c

# The issue is the data set includes regions

# List by claude
non_countries <- c("income", "IDA", "IBRD", "&", "dividend", "small states", "Small states",
                   "Euro area", "European", "World", "OECD", "Indebted", "Fragile",
                   "Least developed", "Arab League", "Baltics", "EASTERN EUROPE",
                   "Sub-Saharan", "SUB-SAHARAN", "South Asia", "North America",
                   "Africa Eastern", "Africa Western", "China and India",
                   "Channel Islands", "Data from", "Last Updated")

diesel_b2016_data_1 <- diesel_b2016_data_1 %>%
  filter(!str_detect(country_name, paste(non_countries, collapse = "|")))

# guess_field(diesel_b2016_data_1$country_code, min_similarity = 50)
# now ~94% match for iso3c, genc3c, etc
# all remaining non-matches are duplicates containing no information

diesel_b2016_data_1 <- diesel_b2016_data_1 %>%
  filter(!country_code %in% c("ADO", "ZAR", "IMY", "WBG", "ROM", "TMP", "KSV")) %>%
  rename(iso3c = country_code)
```

Join

```r
# SELECT relevent variables from transit_data_1
joined <- transit_data_1 %>%
  select(id, iso3c, city, line, phase, rr, length, tunnel, elevated, atgrade,
         stations, platform_length_meters, year, cost_km_2025_dollars)

# ADD relevent variables from legal_origins_data_1
joined <- joined %>%
  left_join(legal_origins_data_1 %>% select( iso3c, legal_systems, mixed_systems_components),
            by = "iso3c")

# ADD relevent variables from colonize_data_1
joined <- joined %>%
  left_join(colonize_data_1 %>% select(iso3c, col_ruler_iso3c),
            by = "iso3c")
```

```
# ADD relevent variables from wb_data_1
joined <- joined %>%
  left_join(wb_data_1 %>% select(-c(country_name, country_code)),
            by = c("iso3c", "year"))

# ADD relevent variables from wage_data_1
joined <- joined %>%
  left_join(wage_data_1 %>% select(iso3c, year, wage_per_month),
            by = c("iso3c", "year"))

# ADD relevent variables from diesel_a2016_data_1
joined <- joined %>%
  left_join(diesel_a2016_data_1 %>% select(iso3c, year, diesel_usd_per_liter_a),
            by = c("iso3c", "year"))

# ADD relevent variables from diesel_a2016_data_1
joined <- joined %>%
  left_join(diesel_b2016_data_1 %>% select(iso3c, year, diesel_usd_per_liter_b),
            by = c("iso3c", "year"))
```

Joined Clean

```
# Make all non-entries into NA using an anonymous function [\(x)]
joined_1 <- joined %>%
  mutate(across(everything(), \(col) if_else(col %in% c(
    "See Notes.", "See notes.", "#N/A", "N/A", "n/a", ".."), NA, col)))

# Clean length:stations and convert to numeric
joined_1 <- joined_1 %>%
  # Remove footnote asterisks
  mutate(across(length:stations, \(col) gsub("\\*", "", col))) %>%
  # Make "x/y" entries NA, get rid of abiguous '36/22' station count
  mutate(across(length:stations, \(col) if_else(grepl("/", col), NA, col))) %>%
  # Convert to numeric
  mutate(across(length:stations, as.numeric))

# Clean platform_length_meters and convert to numeric
joined_1 <- joined_1 %>%
  # Claude code: if m is present, replace value with all digits and '.'s proceeding
  #              m until a non digit/. is found. (ex: '4 cars/120m' -> '120')
```

```
  mutate(platform_length_meters = if_else(
    grepl("m", platform_length_meters),
    sub("^(.*[^0-9.])?([0-9.]+)m.*$", "\\2", platform_length_meters),
    platform_length_meters)) %>%
  # Make any values containing 'c' (cars), '/' or '-' into NA
  mutate(platform_length_meters = if_else(
    grepl("[c/-]", platform_length_meters), NA, platform_length_meters)) %>%
  # Convert to numeric
  mutate(platform_length_meters = as.numeric(platform_length_meters))

# Clean cost_km_2025_dollars and convert to numeric
joined_1 <- joined_1 %>%
  mutate(cost_km_2025_dollars = gsub(",", "", cost_km_2025_dollars)) %>%
  mutate(cost_km_2025_dollars = as.numeric(cost_km_2025_dollars))

# Clean government_effectiveness:trade_percent_of_gdp and convert to numeric
joined_1 <- joined_1 %>%
  # Convert to numeric
  mutate(across(government_effectiveness:trade_percent_of_gdp, as.numeric))

#CHOICE#
# note: split colonial past?
# Correct legal systems for cit(y/ies) that differ from their greater nation
joined_1 <- joined_1 %>%
  mutate(legal_systems = if_else(
    city %in% "Montreal", "Mixed", legal_systems)) %>%
  mutate(mixed_systems_components = if_else(
    city %in% "Montreal", "Civil/Common", mixed_systems_components))

# Unify legal system values format
joined_1 <- joined_1 %>%
  mutate(mixed_systems_components = gsub(" ", "", mixed_systems_components)) %>%
  mutate(mixed_systems_components = gsub("law", "", mixed_systems_components)) %>%
  mutate(legal_systems = gsub(" ", "", legal_systems)) %>%
  mutate(legal_systems = gsub("law", "", legal_systems)) %>%
  mutate(mixed_systems_components = if_else(
    is.na(mixed_systems_components), NA,
    sapply(strsplit(mixed_systems_components, "/"), \(parts) paste(sort(parts), collapse = "/

# create 1 diesel price variale
joined_1 <- joined_1 %>%
  mutate(diesel_usd_per_liter = coalesce(
```

```
    diesel_usd_per_liter_a, diesel_usd_per_liter_b))

# remove opbservations without dependent var values
joined_1 <- joined_1 %>%
  filter(!is.na(cost_km_2025_dollars))
```

Transform Variables

```
# Make variables
joined_2 <- joined_1 %>%
  # Make variables for length of rail type/length of rail
  mutate(tunnel_per = tunnel/length) %>%
  mutate(elevated_per = elevated/length) %>%
  mutate(atgrade_per = atgrade/length) %>%
  # Make variables for stations/km and platform m/km
  mutate(stations_per_km = stations/length) %>%
  mutate(platfrom_m_per_km = stations_per_km*platform_length_meters) %>%
  #CHOICE#
  # Make col_ruler_2 where uncolonized countries are marked as their own col origin
  mutate(col_ruler_2 = if_else(is.na(col_ruler_iso3c), iso3c, col_ruler_iso3c)) %>%
  # Make legal_systems_2, where 'Mixed' _> value of mixed_systems_components
  mutate(legal_systems_2 = if_else(
    legal_systems == "Mixed", mixed_systems_components, legal_systems))


# Remove irrelevent variables
joined_2_trim <- joined_2 %>%
  select(-c(id, tunnel, elevated, atgrade, stations, platform_length_meters,
            legal_systems, mixed_systems_components, col_ruler_iso3c,
            diesel_usd_per_liter_a, diesel_usd_per_liter_b))
```

Data Exploration

Variable Relations

```
num_vars <- joined_2_trim %>% select(where(is.numeric), -cost_km_2025_dollars) %>% names()
cat_vars <- joined_2_trim %>% select(!where(is.numeric)) %>% names()
```

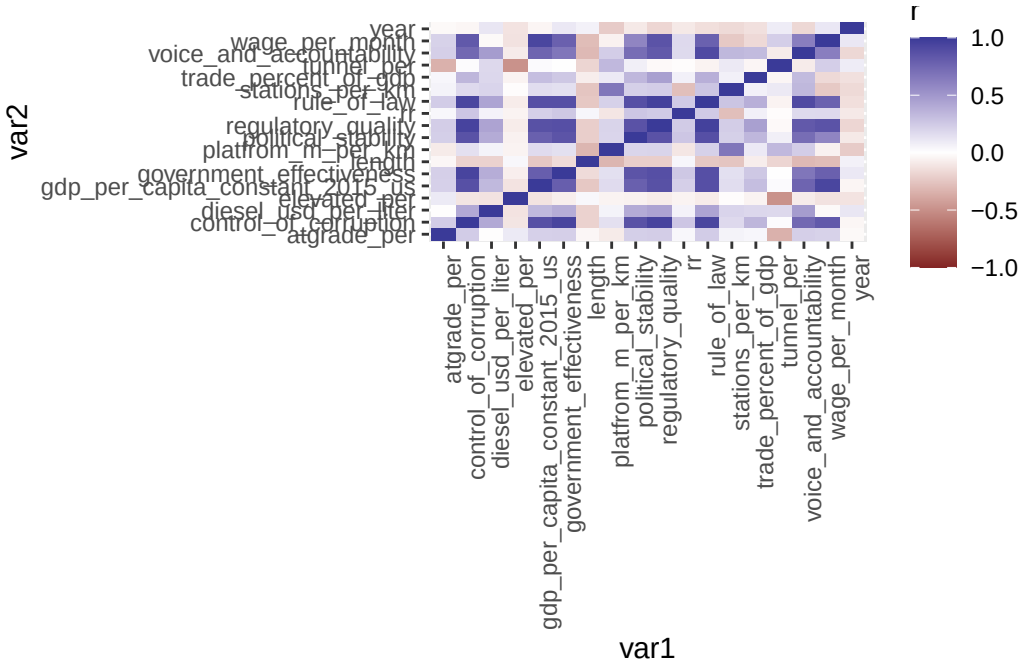
```r
# Step 1: Correlation of each numeric variable with the dependent, ranked by strength
cor_out <- sapply(num_vars, \(v) cor(joined_2_trim[[v]], joined_2_trim$cost_km_2025_dollars,
round(cor_out[order(-abs(cor_out))], 2)
```

```
              platfrom_m_per_km                   wage_per_month
                           0.40                             0.39
gdp_per_capita_constant_2015_us             trade_percent_of_gdp
                           0.27                             0.24
             regulatory_quality                      rule_of_law
                           0.20                             0.19
          control_of_corruption          voice_and_accountability
                           0.18                             0.17
                     tunnel_per          government_effectiveness
                           0.15                             0.13
                         length                               rr
                          -0.12                             0.12
            political_stability                    elevated_per
                           0.11                            -0.11
                    atgrade_per                  stations_per_km
                          -0.07                             0.07
           diesel_usd_per_liter                             year
                          -0.06                            -0.03
```

```r
# Step 2: Correlations among regressors (long format, reused below)
cor_mat <- joined_2_trim %>% select(all_of(num_vars)) %>% cor(use = "pairwise.complete.obs")

cor_long <- cor_mat %>%
  as.data.frame() %>%
  mutate(var1 = rownames(cor_mat)) %>%
  pivot_longer(-var1, names_to = "var2", values_to = "r")

ggplot(cor_long, aes(x = var1, y = var2, fill = r)) +
  geom_tile() +
  scale_fill_gradient2(limits = c(-1, 1)) +
  theme(axis.text.x = element_text(angle = 90, hjust = 1))
```




```
# Step 3: Flag highly correlated regressor pairs (multicollinearity risk)
cor_long %>%
  filter(var1 < var2, abs(r) > 0.8) %>%
  arrange(desc(abs(r)))
```

```
# A tibble: 18 x 3
   var1                            var2                                r
   <chr>                           <chr>                           <dbl>
 1 regulatory_quality              rule_of_law                     0.959
 2 control_of_corruption           regulatory_quality              0.946
 3 control_of_corruption           government_effectiveness        0.944
 4 control_of_corruption           rule_of_law                     0.933
 5 gdp_per_capita_constant_2015_us wage_per_month                  0.932
 6 rule_of_law                     voice_and_accountability        0.907
 7 government_effectiveness        rule_of_law                     0.879
 8 political_stability             rule_of_law                     0.879
 9 gdp_per_capita_constant_2015_us rule_of_law                     0.878
10 government_effectiveness        regulatory_quality              0.878
11 control_of_corruption           gdp_per_capita_constant_2015_us 0.876
12 gdp_per_capita_constant_2015_us regulatory_quality              0.871
13 control_of_corruption           political_stability             0.865
14 political_stability             regulatory_quality              0.862
```

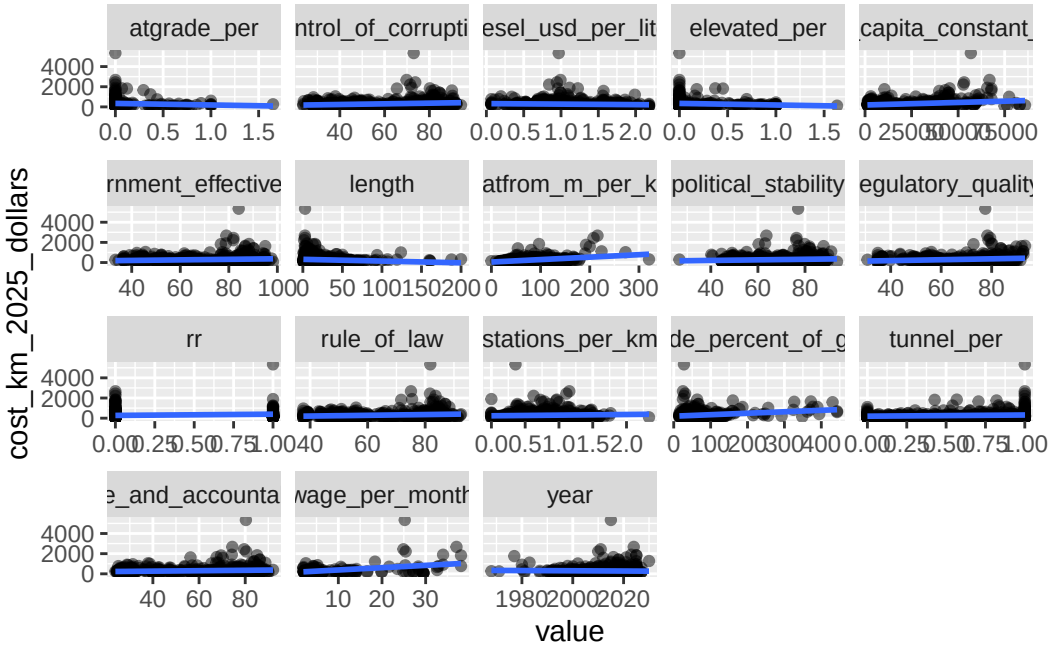
```
15 regulatory_quality              wage_per_month                0.856
16 government_effectiveness        political_stability           0.853
17 control_of_corruption           wage_per_month                0.826
18 regulatory_quality              voice_and_accountability      0.824
```

```r
# Step 4: Outcome vs. every numeric variable, one faceted plot
joined_2_trim %>%
  select(cost_km_2025_dollars, all_of(num_vars)) %>%
  pivot_longer(-cost_km_2025_dollars, names_to = "var", values_to = "value") %>%
  ggplot(aes(x = value, y = cost_km_2025_dollars)) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm", se = FALSE) +
  facet_wrap(~ var, scales = "free_x")
```

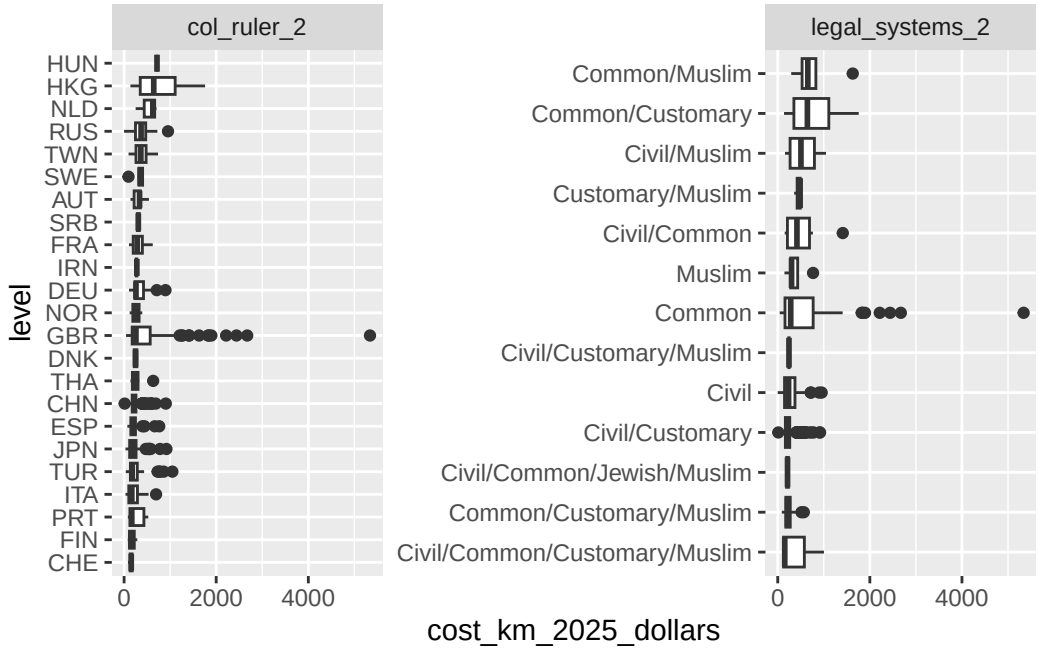
```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 3325 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 3325 rows containing missing values or values outside the scale range
(`geom_point()`).
```

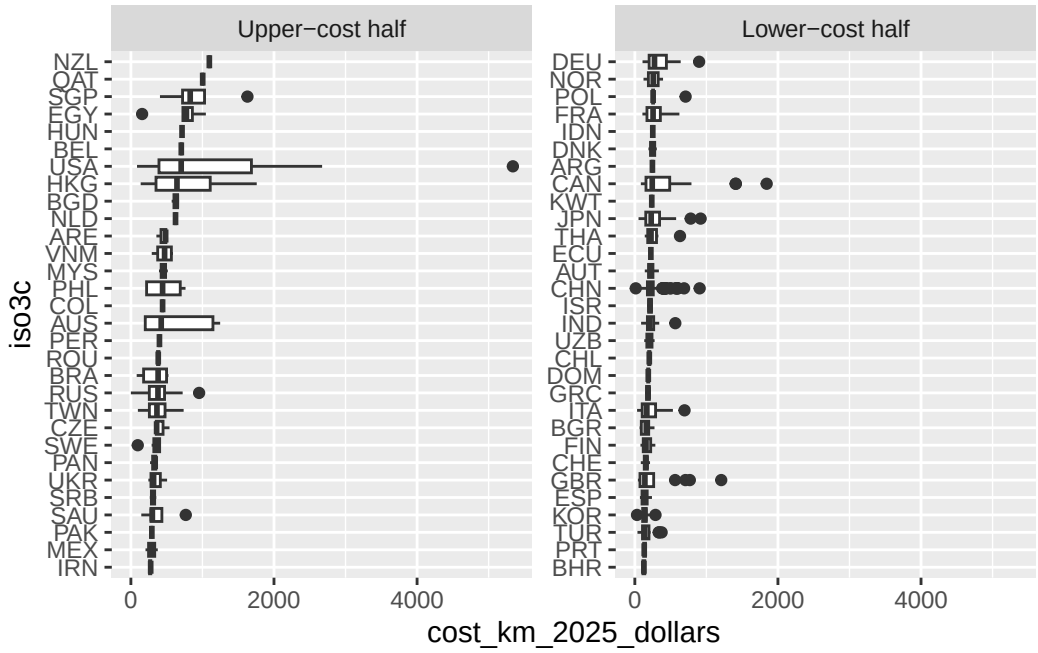


```r
# Step 5: Outcome by every (relevent) categorical variable
  # col_ruler_2 and legal_systems_2
joined_2_trim %>%
  select(cost_km_2025_dollars, all_of(cat_vars)) %>%
  select(-c(city, line, phase, iso3c)) %>%
  pivot_longer(-cost_km_2025_dollars, names_to = "var", values_to = "level",
               values_transform = as.character) %>%
  ggplot(aes(x = cost_km_2025_dollars,
             y = reorder(level, cost_km_2025_dollars, median, na.rm = TRUE))) +
  geom_boxplot() +
  facet_wrap(~ var, scales = "free_y") +
  labs(y = "level")
```




```r
  # by country
joined_2_trim %>%
  group_by(iso3c) %>%
  mutate(med = median(cost_km_2025_dollars, na.rm = TRUE)) %>%
  ungroup() %>%
  mutate(half = factor(if_else(dense_rank(med) <= n_distinct(iso3c) / 2,
                               "Lower-cost half", "Upper-cost half"),
                       levels = c("Upper-cost half", "Lower-cost half"))) %>%
  ggplot(aes(x = cost_km_2025_dollars,
```

```
            y = reorder(iso3c, cost_km_2025_dollars, median, na.rm = TRUE))) +
  geom_boxplot() +
  facet_wrap(~ half, scales = "free_y") +
  labs(y = "iso3c")
```




```
# Step 6: Bivariate regression for each numeric variable (slope and p-value)
biv <- t(sapply(num_vars, \(v)
                summary(lm(reformulate(v, "cost_km_2025_dollars"), data = joined_2_trim))$co
round(biv, 3)
```

```
                                Estimate Pr(>|t|)
rr                               121.926    0.000
length                            -1.716    0.000
year                              -1.168    0.271
government_effectiveness           2.730    0.000
control_of_corruption              3.323    0.000
political_stability                2.866    0.000
regulatory_quality                 4.448    0.000
rule_of_law                        3.706    0.000
voice_and_accountability           2.092    0.000
gdp_per_capita_constant_2015_us    0.005    0.000
trade_percent_of_gdp               1.459    0.000
```

```
wage_per_month                  22.803    0.000
diesel_usd_per_liter           -51.336    0.042
tunnel_per                     119.335    0.000
elevated_per                  -157.201    0.005
atgrade_per                   -153.701    0.063
stations_per_km                 64.527    0.019
platfrom_m_per_km                2.430    0.000
```

Variable Relations (no outlier)

```
joined_2_trim_no_max <- joined_2_trim %>%
  filter(row_number() != which.max(cost_km_2025_dollars))

num_vars <- joined_2_trim_no_max %>% select(where(is.numeric), -cost_km_2025_dollars) %>% nar
cat_vars <- joined_2_trim_no_max %>% select(!where(is.numeric)) %>% names()

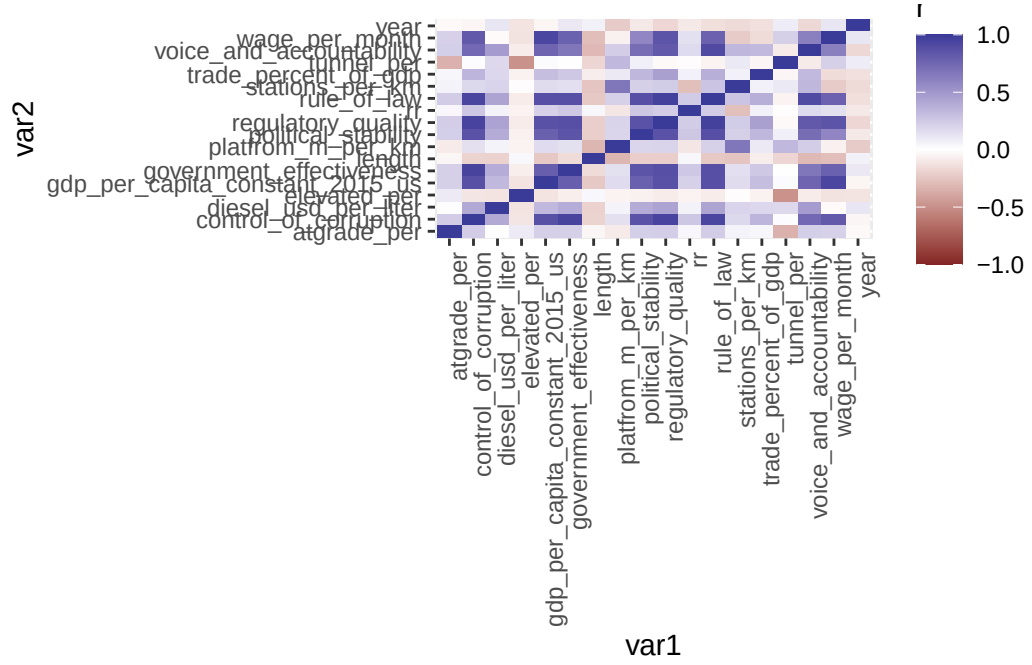
# Step 1: Correlation of each numeric variable with the dependent, ranked by strength
cor_out <- sapply(num_vars, \(v) cor(joined_2_trim_no_max[[v]], joined_2_trim_no_max$cost_km_
round(cor_out[order(-abs(cor_out))], 2)
```

```
             wage_per_month              platfrom_m_per_km
                       0.43                           0.40
       trade_percent_of_gdp gdp_per_capita_constant_2015_us
                       0.29                           0.28
         regulatory_quality                    rule_of_law
                       0.21                           0.19
    control_of_corruption       voice_and_accountability
                       0.19                           0.17
                tunnel_per                         length
                       0.16                          -0.13
  government_effectiveness                   elevated_per
                       0.13                          -0.12
       political_stability               stations_per_km
                       0.11                           0.10
                atgrade_per                             rr
                      -0.08                           0.08
      diesel_usd_per_liter                           year
                      -0.07                          -0.04
```

```r
# Step 2: Correlations among regressors (long format, reused below)
cor_mat <- joined_2_trim_no_max %>% select(all_of(num_vars)) %>% cor(use = "pairwise.complete

cor_long <- cor_mat %>%
  as.data.frame() %>%
  mutate(var1 = rownames(cor_mat)) %>%
  pivot_longer(-var1, names_to = "var2", values_to = "r")

ggplot(cor_long, aes(x = var1, y = var2, fill = r)) +
  geom_tile() +
  scale_fill_gradient2(limits = c(-1, 1)) +
  theme(axis.text.x = element_text(angle = 90, hjust = 1))
```




```r
# Step 3: Flag highly correlated regressor pairs (multicollinearity risk)
cor_long %>%
  filter(var1 < var2, abs(r) > 0.8) %>%
  arrange(desc(abs(r)))
```

```
# A tibble: 18 x 3
   var1                  var2                      r
   <chr>                 <chr>                 <dbl>
 1 regulatory_quality    rule_of_law           0.959
```

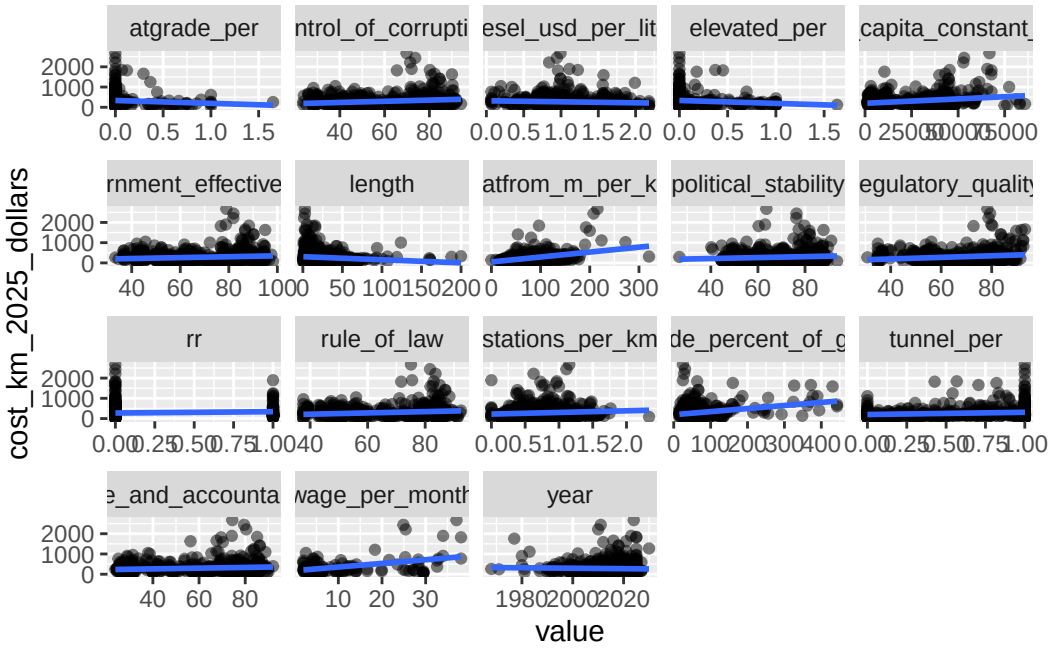
```
 2 control_of_corruption           regulatory_quality              0.946
 3 control_of_corruption           government_effectiveness        0.944
 4 control_of_corruption           rule_of_law                     0.933
 5 gdp_per_capita_constant_2015_us wage_per_month                  0.931
 6 rule_of_law                     voice_and_accountability        0.907
 7 political_stability             rule_of_law                     0.878
 8 government_effectiveness        rule_of_law                     0.878
 9 gdp_per_capita_constant_2015_us rule_of_law                     0.878
10 government_effectiveness        regulatory_quality              0.877
11 control_of_corruption           gdp_per_capita_constant_2015_us 0.876
12 gdp_per_capita_constant_2015_us regulatory_quality              0.871
13 control_of_corruption           political_stability             0.865
14 political_stability             regulatory_quality              0.862
15 regulatory_quality              wage_per_month                  0.854
16 government_effectiveness        political_stability             0.853
17 regulatory_quality              voice_and_accountability        0.824
18 control_of_corruption           wage_per_month                  0.824
```

```r
# Step 4: Outcome vs. every numeric variable, one faceted plot
joined_2_trim_no_max %>%
  select(cost_km_2025_dollars, all_of(num_vars)) %>%
  pivot_longer(-cost_km_2025_dollars, names_to = "var", values_to = "value") %>%
  ggplot(aes(x = value, y = cost_km_2025_dollars)) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm", se = FALSE) +
  facet_wrap(~ var, scales = "free_x")
```

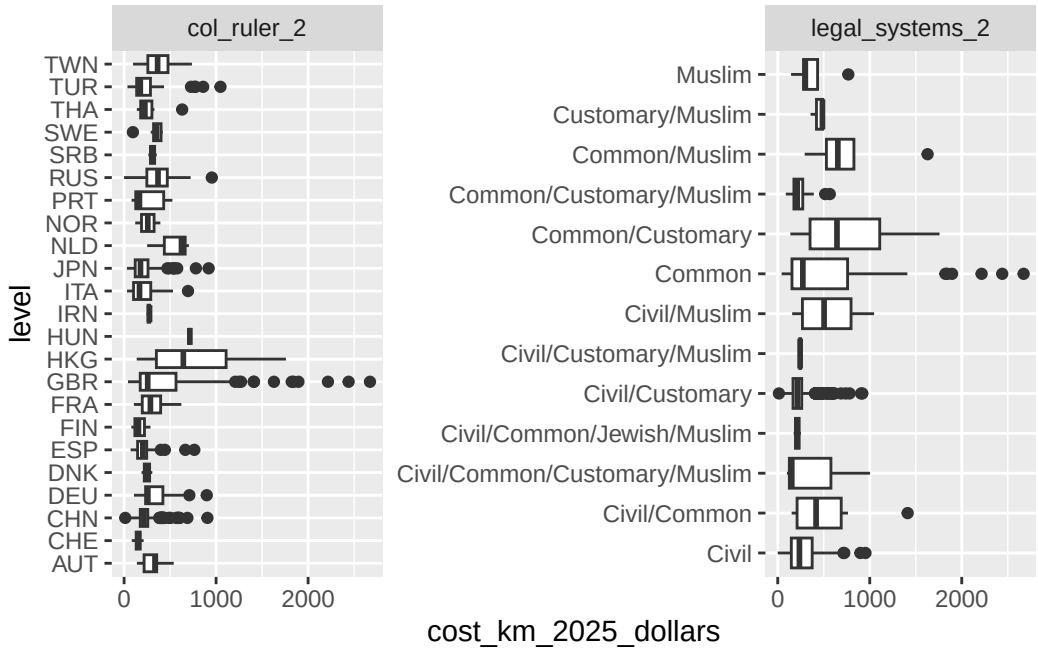
```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 3324 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

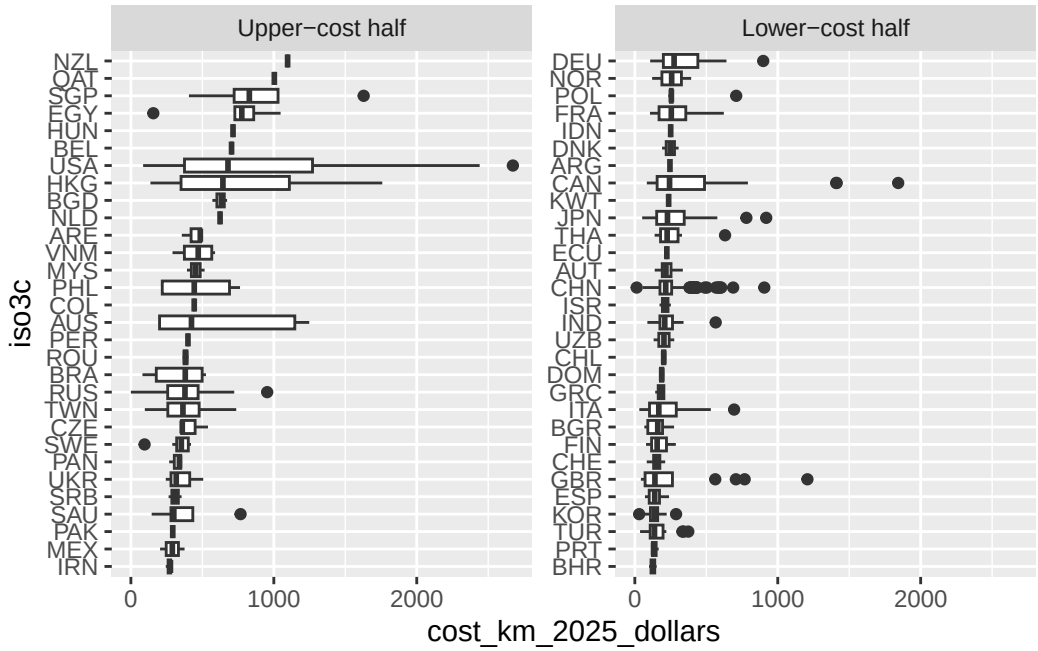
```
Warning: Removed 3324 rows containing missing values or values outside the scale range
(`geom_point()`).
```




```
# Step 5: Outcome by every (relevent) categorical variable
  # col_ruler_2 and legal_systems_2
joined_2_trim_no_max %>%
  select(cost_km_2025_dollars, all_of(cat_vars)) %>%
  select(-c(city, line, phase, iso3c)) %>%
  pivot_longer(-cost_km_2025_dollars, names_to = "var", values_to = "level",
               values_transform = as.character) %>%
  ggplot(aes(x = cost_km_2025_dollars, y = level)) +
  geom_boxplot() +
  facet_wrap(~ var, scales = "free_y")
```




```r
  # by country
joined_2_trim_no_max %>%
  group_by(iso3c) %>%
  mutate(med = median(cost_km_2025_dollars, na.rm = TRUE)) %>%
  ungroup() %>%
  mutate(half = factor(if_else(dense_rank(med) <= n_distinct(iso3c) / 2,
                               "Lower-cost half", "Upper-cost half"),
                       levels = c("Upper-cost half", "Lower-cost half"))) %>%
  ggplot(aes(x = cost_km_2025_dollars,
             y = reorder(iso3c, cost_km_2025_dollars, median, na.rm = TRUE))) +
  geom_boxplot() +
  facet_wrap(~ half, scales = "free_y") +
  labs(y = "iso3c")
```




```
# Step 6: Bivariate regression for each numeric variable (slope and p-value)
biv <- t(sapply(num_vars, \(v)
                summary(lm(reformulate(v, "cost_km_2025_dollars"), data = joined_2_trim_no_ma
round(biv, 3)
```

	Estimate	Pr(>|t|)
rr	66.380	0.012
length	-1.528	0.000
year	-1.085	0.226
government_effectiveness	2.243	0.000
control_of_corruption	2.907	0.000
political_stability	2.370	0.000
regulatory_quality	3.916	0.000
rule_of_law	3.169	0.000
voice_and_accountability	1.785	0.000
gdp_per_capita_constant_2015_us	0.004	0.000
trade_percent_of_gdp	1.519	0.000
wage_per_month	18.101	0.000
diesel_usd_per_liter	-49.004	0.018
tunnel_per	110.621	0.000
elevated_per	-144.930	0.002
atgrade_per	-141.344	0.039

```
stations_per_km                  81.220    0.000
platfrom_m_per_km                 2.430    0.000
```

```r
# create correlation matrix
cor_matrix <- joined_2_trim %>%
  select(where(is.numeric)) %>%
  cor(use = "pairwise.complete.obs")

cor_round <- round(cor_matrix, 2)
colnames(cor_round) <- abbreviate(colnames(cor_round), 4)
cor_round
```

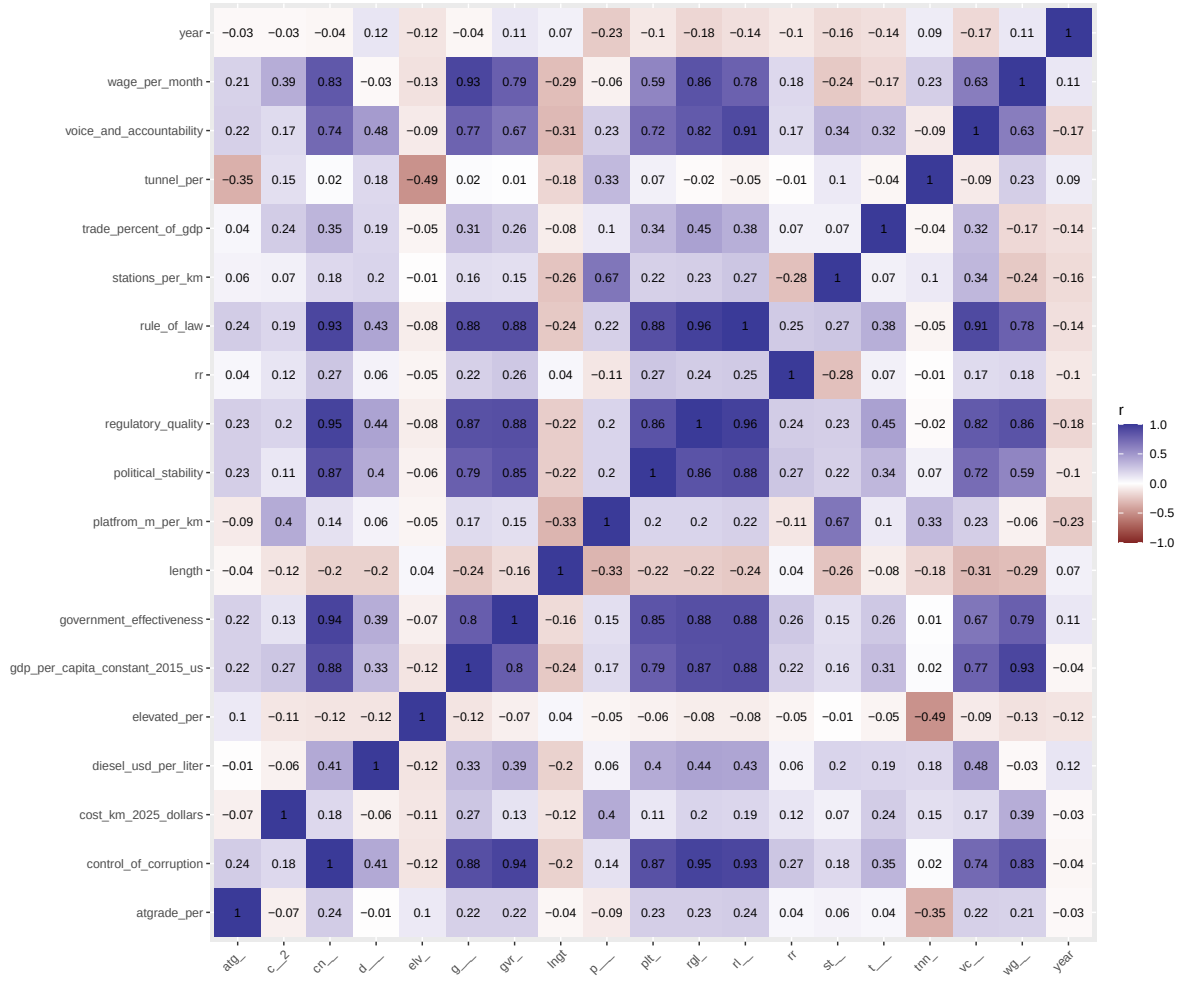
```
                                  rr  lngt  year  c__2  gvr_  cn__  plt_  rgl_
rr                              1.00  0.04 -0.10  0.12  0.26  0.27  0.27  0.24
length                          0.04  1.00  0.07 -0.12 -0.16 -0.20 -0.22 -
0.22
year                           -0.10  0.07  1.00 -0.03  0.11 -0.04 -0.10 -
0.18
cost_km_2025_dollars            0.12 -0.12 -0.03  1.00  0.13  0.18  0.11  0.20
government_effectiveness        0.26 -0.16  0.11  0.13  1.00  0.94  0.85  0.88
control_of_corruption           0.27 -0.20 -0.04  0.18  0.94  1.00  0.87  0.95
political_stability             0.27 -0.22 -0.10  0.11  0.85  0.87  1.00  0.86
regulatory_quality              0.24 -0.22 -0.18  0.20  0.88  0.95  0.86  1.00
rule_of_law                     0.25 -0.24 -0.14  0.19  0.88  0.93  0.88  0.96
voice_and_accountability        0.17 -0.31 -0.17  0.17  0.67  0.74  0.72  0.82
gdp_per_capita_constant_2015_us 0.22 -0.24 -0.04  0.27  0.80  0.88  0.79  0.87
trade_percent_of_gdp            0.07 -0.08 -0.14  0.24  0.26  0.35  0.34  0.45
wage_per_month                  0.18 -0.29  0.11  0.39  0.79  0.83  0.59  0.86
diesel_usd_per_liter            0.06 -0.20  0.12 -0.06  0.39  0.41  0.40  0.44
tunnel_per                     -0.01 -0.18  0.09  0.15  0.01  0.02  0.07 -
0.02
elevated_per                   -0.05  0.04 -0.12 -0.11 -0.07 -0.12 -0.06 -
0.08
atgrade_per                     0.04 -0.04 -0.03 -0.07  0.22  0.24  0.23  0.23
stations_per_km                -0.28 -0.26 -0.16  0.07  0.15  0.18  0.22  0.23
platfrom_m_per_km              -0.11 -0.33 -0.23  0.40  0.15  0.14  0.20  0.20
                                rl__  vc__  g___  t___  wg__  d___  tnn_  elv_
rr                              0.25  0.17  0.22  0.07  0.18  0.06 -0.01 -
0.05
length                         -0.24 -0.31 -0.24 -0.08 -0.29 -0.20 -0.18  0.04
year                           -0.14 -0.17 -0.04 -0.14  0.11  0.12  0.09 -
0.12
```

```
cost_km_2025_dollars             0.19  0.17  0.27  0.24  0.39 -0.06  0.15 -
0.11
government_effectiveness         0.88  0.67  0.80  0.26  0.79  0.39  0.01 -
0.07
control_of_corruption            0.93  0.74  0.88  0.35  0.83  0.41  0.02 -
0.12
political_stability              0.88  0.72  0.79  0.34  0.59  0.40  0.07 -
0.06
regulatory_quality               0.96  0.82  0.87  0.45  0.86  0.44 -0.02 -
0.08
rule_of_law                      1.00  0.91  0.88  0.38  0.78  0.43 -0.05 -
0.08
voice_and_accountability         0.91  1.00  0.77  0.32  0.63  0.48 -0.09 -
0.09
gdp_per_capita_constant_2015_us  0.88  0.77  1.00  0.31  0.93  0.33  0.02 -
0.12
trade_percent_of_gdp             0.38  0.32  0.31  1.00 -0.17  0.19 -0.04 -
0.05
wage_per_month                   0.78  0.63  0.93 -0.17  1.00 -0.03  0.23 -
0.13
diesel_usd_per_liter             0.43  0.48  0.33  0.19 -0.03  1.00  0.18 -
0.12
tunnel_per                      -0.05 -0.09  0.02 -0.04  0.23  0.18  1.00 -
0.49
elevated_per                    -0.08 -0.09 -0.12 -0.05 -0.13 -0.12 -0.49  1.00
atgrade_per                      0.24  0.22  0.22  0.04  0.21 -0.01 -0.35  0.10
stations_per_km                  0.27  0.34  0.16  0.07 -0.24  0.20  0.10 -
0.01
platfrom_m_per_km                0.22  0.23  0.17  0.10 -0.06  0.06  0.33 -
0.05

                                 atg_  st__  p___
rr                               0.04 -0.28 -0.11
length                          -0.04 -0.26 -0.33
year                            -0.03 -0.16 -0.23
cost_km_2025_dollars            -0.07  0.07  0.40
government_effectiveness         0.22  0.15  0.15
control_of_corruption            0.24  0.18  0.14
political_stability              0.23  0.22  0.20
regulatory_quality               0.23  0.23  0.20
rule_of_law                      0.24  0.27  0.22
voice_and_accountability         0.22  0.34  0.23
gdp_per_capita_constant_2015_us  0.22  0.16  0.17
trade_percent_of_gdp             0.04  0.07  0.10
```

```
wage_per_month                0.21 -0.24 -0.06
diesel_usd_per_liter         -0.01  0.20  0.06
tunnel_per                   -0.35  0.10  0.33
elevated_per                  0.10 -0.01 -0.05
atgrade_per                   1.00  0.06 -0.09
stations_per_km               0.06  1.00  0.67
platfrom_m_per_km            -0.09  0.67  1.00
```

```r
cor_round %>%
  as.data.frame() %>%
  rownames_to_column("var1") %>%
  pivot_longer(-var1, names_to = "var2", values_to = "r") %>%
  ggplot(aes(var2, var1, fill = r)) +
  geom_tile() +
  geom_text(aes(label = r), size = 3) +
  scale_fill_gradient2(limits = c(-1, 1)) +
  labs(x = NULL, y = NULL, fill = "r") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```




```
# create a table from the matrix and sort by absolute value of correlation
cor_matrix %>%
  as.table() %>%
  as.data.frame() %>%
  filter(as.character(Var1) < as.character(Var2)) %>%
  rename(cor = Freq) %>%
  arrange(desc(abs(cor)))
```

	Var1	Var2
1	regulatory_quality	rule_of_law
2	control_of_corruption	regulatory_quality
3	control_of_corruption	government_effectiveness
4	control_of_corruption	rule_of_law
5	gdp_per_capita_constant_2015_us	wage_per_month

6	rule_of_law	voice_and_accountability
7	government_effectiveness	rule_of_law
8	political_stability	rule_of_law
9	gdp_per_capita_constant_2015_us	rule_of_law
10	government_effectiveness	regulatory_quality
11	control_of_corruption	gdp_per_capita_constant_2015_us
12	gdp_per_capita_constant_2015_us	regulatory_quality
13	control_of_corruption	political_stability
14	political_stability	regulatory_quality
15	regulatory_quality	wage_per_month
16	government_effectiveness	political_stability
17	control_of_corruption	wage_per_month
18	regulatory_quality	voice_and_accountability
19	gdp_per_capita_constant_2015_us	government_effectiveness
20	gdp_per_capita_constant_2015_us	political_stability
21	government_effectiveness	wage_per_month
22	rule_of_law	wage_per_month
23	gdp_per_capita_constant_2015_us	voice_and_accountability
24	control_of_corruption	voice_and_accountability
25	political_stability	voice_and_accountability
26	platfrom_m_per_km	stations_per_km
27	government_effectiveness	voice_and_accountability
28	voice_and_accountability	wage_per_month
29	political_stability	wage_per_month
30	elevated_per	tunnel_per
31	diesel_usd_per_liter	voice_and_accountability
32	regulatory_quality	trade_percent_of_gdp
33	diesel_usd_per_liter	regulatory_quality
34	diesel_usd_per_liter	rule_of_law
35	control_of_corruption	diesel_usd_per_liter
36	cost_km_2025_dollars	platfrom_m_per_km
37	diesel_usd_per_liter	political_stability
38	diesel_usd_per_liter	government_effectiveness
39	cost_km_2025_dollars	wage_per_month
40	rule_of_law	trade_percent_of_gdp
41	atgrade_per	tunnel_per
42	control_of_corruption	trade_percent_of_gdp
43	political_stability	trade_percent_of_gdp
44	stations_per_km	voice_and_accountability
45	platfrom_m_per_km	tunnel_per
46	diesel_usd_per_liter	gdp_per_capita_constant_2015_us
47	length	platfrom_m_per_km
48	trade_percent_of_gdp	voice_and_accountability

49	length	voice_and_accountability
50	gdp_per_capita_constant_2015_us	trade_percent_of_gdp
51	length	wage_per_month
52	rr	stations_per_km
53	rule_of_law	stations_per_km
54	cost_km_2025_dollars	gdp_per_capita_constant_2015_us
55	political_stability	rr
56	control_of_corruption	rr
57	government_effectiveness	rr
58	government_effectiveness	trade_percent_of_gdp
59	length	stations_per_km
60	rr	rule_of_law
61	regulatory_quality	rr
62	gdp_per_capita_constant_2015_us	length
63	length	rule_of_law
64	stations_per_km	wage_per_month
65	atgrade_per	rule_of_law
66	atgrade_per	control_of_corruption
67	cost_km_2025_dollars	trade_percent_of_gdp
68	platfrom_m_per_km	voice_and_accountability
69	regulatory_quality	stations_per_km
70	atgrade_per	regulatory_quality
71	tunnel_per	wage_per_month
72	platfrom_m_per_km	year
73	atgrade_per	political_stability
74	atgrade_per	voice_and_accountability
75	gdp_per_capita_constant_2015_us	rr
76	atgrade_per	government_effectiveness
77	political_stability	stations_per_km
78	atgrade_per	gdp_per_capita_constant_2015_us
79	length	political_stability
80	platfrom_m_per_km	rule_of_law
81	length	regulatory_quality
82	atgrade_per	wage_per_month
83	platfrom_m_per_km	regulatory_quality
84	diesel_usd_per_liter	stations_per_km
85	diesel_usd_per_liter	length
86	cost_km_2025_dollars	regulatory_quality
87	control_of_corruption	length
88	platfrom_m_per_km	political_stability
89	cost_km_2025_dollars	rule_of_law
90	diesel_usd_per_liter	trade_percent_of_gdp
91	length	tunnel_per

92	diesel_usd_per_liter	tunnel_per
93	control_of_corruption	cost_km_2025_dollars
94	regulatory_quality	year
95	control_of_corruption	stations_per_km
96	rr	wage_per_month
97	voice_and_accountability	year
98	rr	voice_and_accountability
99	cost_km_2025_dollars	voice_and_accountability
100	gdp_per_capita_constant_2015_us	platfrom_m_per_km
101	trade_percent_of_gdp	wage_per_month
102	stations_per_km	year
103	gdp_per_capita_constant_2015_us	stations_per_km
104	government_effectiveness	length
105	cost_km_2025_dollars	tunnel_per
106	government_effectiveness	platfrom_m_per_km
107	government_effectiveness	stations_per_km
108	control_of_corruption	platfrom_m_per_km
109	rule_of_law	year
110	trade_percent_of_gdp	year
111	cost_km_2025_dollars	government_effectiveness
112	elevated_per	wage_per_month
113	cost_km_2025_dollars	length
114	diesel_usd_per_liter	elevated_per
115	elevated_per	gdp_per_capita_constant_2015_us
116	elevated_per	year
117	cost_km_2025_dollars	rr
118	diesel_usd_per_liter	year
119	control_of_corruption	elevated_per
120	cost_km_2025_dollars	political_stability
121	cost_km_2025_dollars	elevated_per
122	wage_per_month	year
123	platfrom_m_per_km	rr
124	government_effectiveness	year
125	stations_per_km	tunnel_per
126	platfrom_m_per_km	trade_percent_of_gdp
127	atgrade_per	elevated_per
128	political_stability	year
129	rr	year
130	tunnel_per	year
131	atgrade_per	platfrom_m_per_km
132	tunnel_per	voice_and_accountability
133	elevated_per	voice_and_accountability
134	elevated_per	regulatory_quality

135	elevated_per	rule_of_law
136	length	trade_percent_of_gdp
137	political_stability	tunnel_per
138	atgrade_per	cost_km_2025_dollars
139	elevated_per	government_effectiveness
140	stations_per_km	trade_percent_of_gdp
141	cost_km_2025_dollars	stations_per_km
142	rr	trade_percent_of_gdp
143	length	year
144	cost_km_2025_dollars	diesel_usd_per_liter
145	elevated_per	political_stability
146	platfrom_m_per_km	wage_per_month
147	diesel_usd_per_liter	rr
148	atgrade_per	stations_per_km
149	diesel_usd_per_liter	platfrom_m_per_km
150	elevated_per	trade_percent_of_gdp
151	elevated_per	platfrom_m_per_km
152	elevated_per	rr
153	rule_of_law	tunnel_per
154	atgrade_per	trade_percent_of_gdp
155	gdp_per_capita_constant_2015_us	year
156	trade_percent_of_gdp	tunnel_per
157	control_of_corruption	year
158	length	rr
159	atgrade_per	length
160	atgrade_per	rr
161	elevated_per	length
162	cost_km_2025_dollars	year
163	atgrade_per	year
164	diesel_usd_per_liter	wage_per_month
165	regulatory_quality	tunnel_per
166	gdp_per_capita_constant_2015_us	tunnel_per
167	control_of_corruption	tunnel_per
168	elevated_per	stations_per_km
169	rr	tunnel_per
170	atgrade_per	diesel_usd_per_liter
171	government_effectiveness	tunnel_per

	cor
1	0.958803892
2	0.946393292
3	0.943819303
4	0.933033748
5	0.931869751

```
6    0.907147297
7    0.878683151
8    0.878577621
9    0.878294265
10   0.877608188
11   0.875890115
12   0.871041859
13   0.865249309
14   0.861903649
15   0.855757603
16   0.853282636
17   0.825620714
18   0.824183141
19   0.799254552
20   0.788153214
21   0.785371893
22   0.782753620
23   0.765371317
24   0.744561245
25   0.717218467
26   0.670864480
27   0.670105064
28   0.625216703
29   0.590605979
30  -0.492328763
31   0.475794519
32   0.454820765
33   0.436852398
34   0.433128840
35   0.408606270
36   0.399662590
37   0.397916546
38   0.389838583
39   0.388883846
40   0.377798440
41  -0.347942708
42   0.347171426
43   0.339572187
44   0.337924771
45   0.333791348
46   0.332057536
47  -0.327064271
48   0.322476585
```

```
49  -0.309810443
50   0.305912197
51  -0.293539323
52  -0.284092932
53   0.273915324
54   0.273626701
55   0.266966640
56   0.266334324
57   0.259994971
58   0.259498789
59  -0.255852152
60   0.247457122
61   0.243872811
62  -0.243381567
63  -0.242129838
64  -0.239565071
65   0.237110689
66   0.237034368
67   0.236147598
68   0.234736587
69   0.234193395
70   0.232146039
71   0.229326207
72  -0.228257704
73   0.226001580
74   0.224896530
75   0.224397210
76   0.223127714
77   0.220565984
78   0.219627696
79  -0.215415858
80   0.215374783
81  -0.215005897
82   0.212434669
83   0.202957184
84   0.199440503
85  -0.196946802
86   0.196771889
87  -0.196387840
88   0.196075416
89   0.188505373
90   0.188086537
91  -0.184133235
```

```
92   0.181467572
93   0.180975694
94  -0.178768405
95   0.176666956
96   0.175915984
97  -0.173733060
98   0.173469855
99   0.171019246
100  0.166873780
101 -0.165670870
102 -0.164425061
103  0.158930714
104 -0.156366976
105  0.148742830
106  0.146414744
107  0.145940452
108  0.143960216
109 -0.142454383
110 -0.135385155
111  0.131237080
112 -0.127914690
113 -0.124579743
114 -0.123710751
115 -0.122489896
116 -0.121045002
117  0.117457643
118  0.116217538
119 -0.115965828
120  0.112962366
121 -0.110847045
122  0.109466996
123 -0.108292017
124  0.106373763
125  0.103148040
126  0.100364348
127  0.097965654
128 -0.097165119
129 -0.096366111
130  0.090092817
131 -0.087932049
132 -0.086979618
133 -0.085370896
134 -0.084459490
```

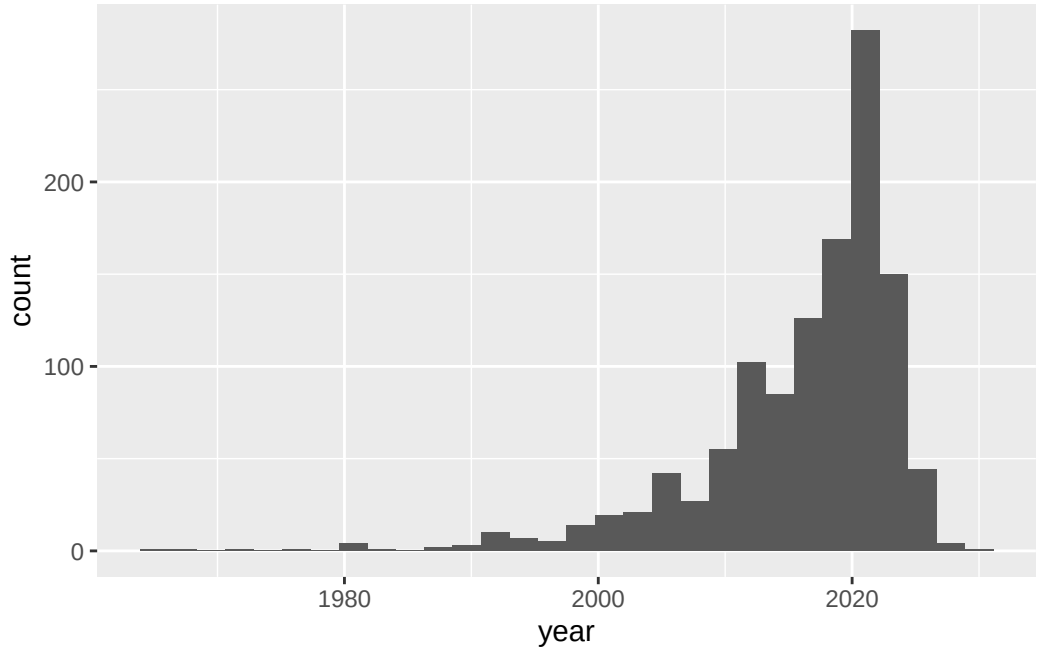
```
135 -0.084160340
136 -0.078803707
137  0.074854367
138 -0.074575470
139 -0.072308822
140  0.070861279
141  0.069908937
142  0.068643843
143  0.065640468
144 -0.062137301
145 -0.062033861
146 -0.061807593
147  0.060224278
148  0.057315904
149  0.057005256
150 -0.054498058
151 -0.054031779
152 -0.052425576
153 -0.052053952
154  0.043550812
155 -0.043451893
156 -0.041817620
157 -0.041123923
158  0.039766227
159 -0.037840668
160  0.037696348
161  0.036075725
162 -0.032438493
163 -0.027953790
164 -0.027016158
165 -0.024284498
166  0.022982767
167  0.018267789
168 -0.012473719
169 -0.011873898
170 -0.007398205
171  0.005584654
```

Year Distribution

```r
ggplot(transit_data_1, aes(x = year)) +
  geom_histogram()
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

```
Warning: Removed 37 rows containing non-finite outside the scale range
(`stat_bin()`).
```

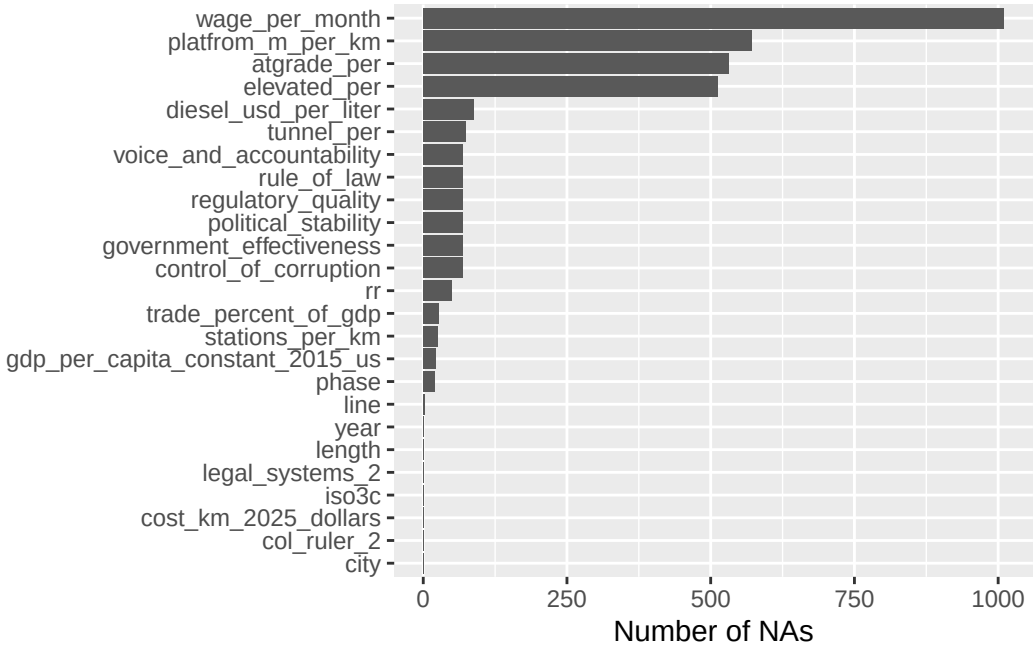



```r
# ggplot(wage_data_1, aes(x = year)) +
#   geom_histogram()
```

NA Values by Variable

```r
# NAs per variable
joined_2_trim %>%
  is.na() %>%
  colSums() %>%
  enframe(name = "variable", value = "n_na") %>%
```

```
  ggplot(aes(x = n_na, y = reorder(variable, n_na))) +
  geom_col() +
  labs(x = "Number of NAs", y = NULL)
```




```
joined_2_trim %>%
  nrow()
```

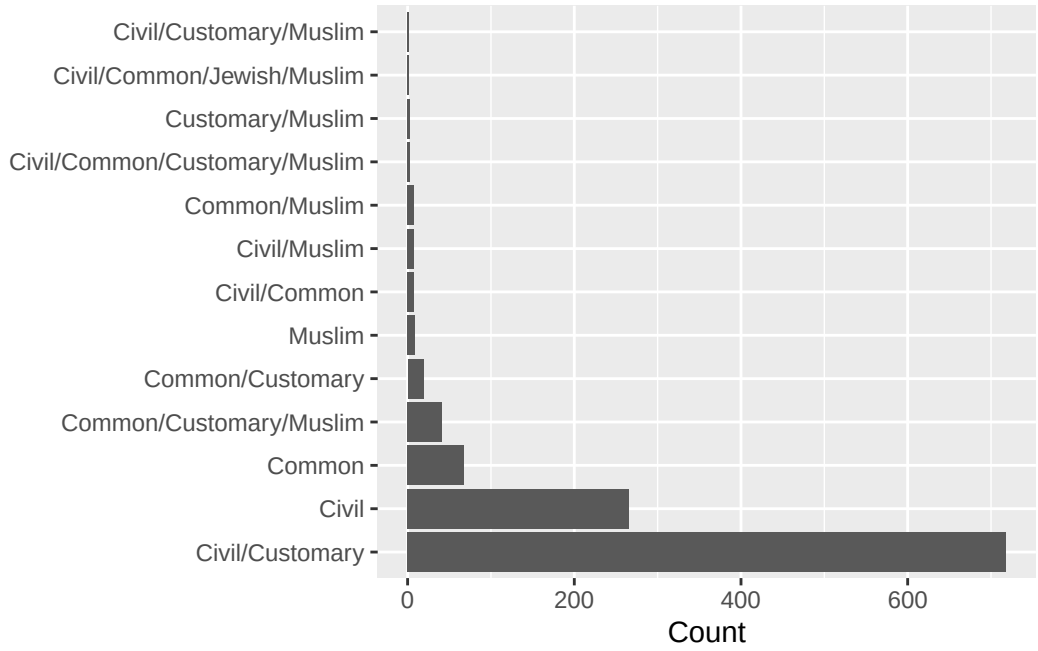
```
[1] 1155
```

```
#CHOICE#NOTACTUALLYTHOMORELIKE#CRITICAL#
joined_2_trim %>%
  filter(if_all(
    -c(platfrom_m_per_km, atgrade_per, elevated_per), \(col) !is.na(col))) %>%
  nrow()
```

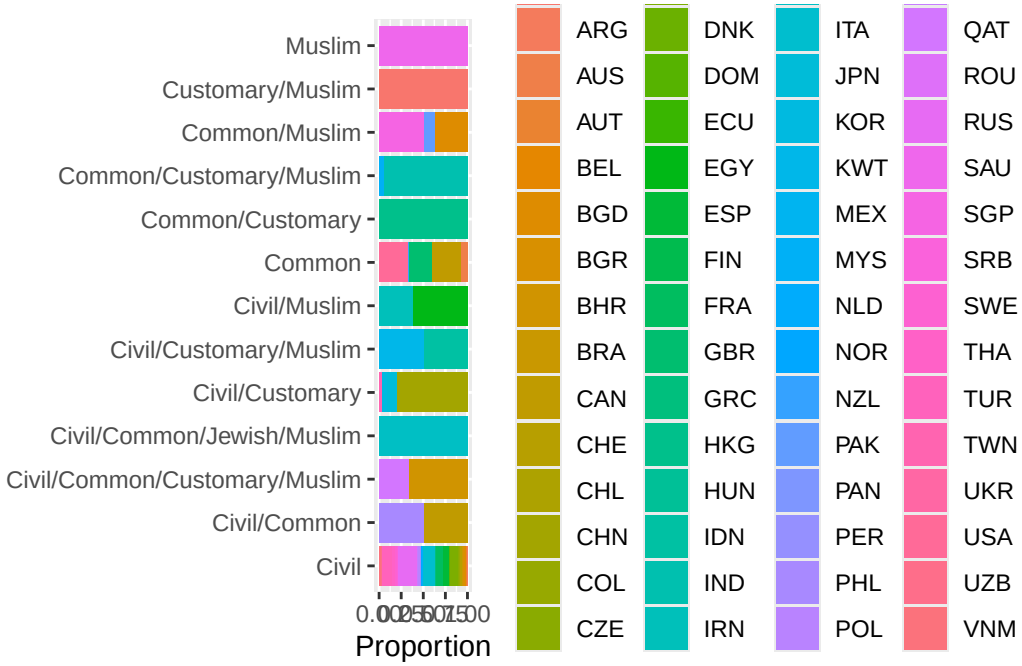
```
[1] 134
```

Legal system and Countries

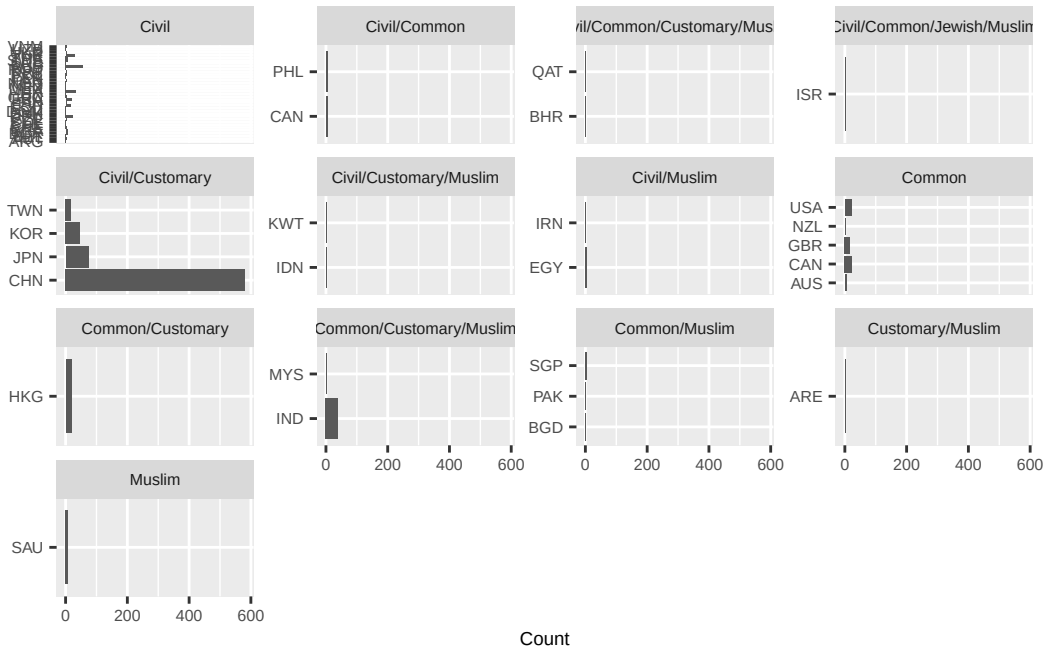
```r
# Distribution barchart of legal_systems_2
joined_2 %>%
  ggplot(aes(y = fct_infreq(legal_systems_2))) +
  geom_bar() +
  labs(x = "Count", y = NULL)
```



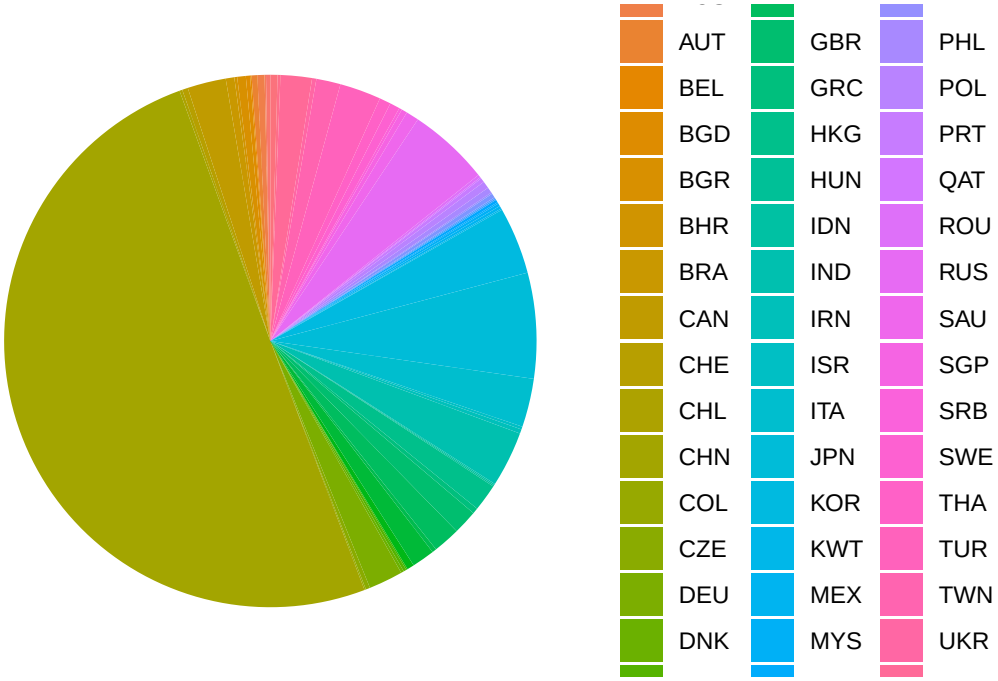

```r
# Distribution of countries, grouped by legal_systems_2
joined_2 %>%
  ggplot(aes(y = legal_systems_2, fill = iso3c)) +
  geom_bar(position = "fill") +
  labs(x = "Proportion", y = NULL, fill = NULL) +
  guides(fill = guide_legend(ncol = 4))
```




```r
# Country distribution by legal system
joined_2 %>%
  ggplot(aes(y = iso3c)) +
  geom_bar() +
  facet_wrap(~ legal_systems_2, scales = "free_y", ncol = 4) +
  labs(x = "Count", y = NULL) +
  theme(text = element_text(size = 7))
```



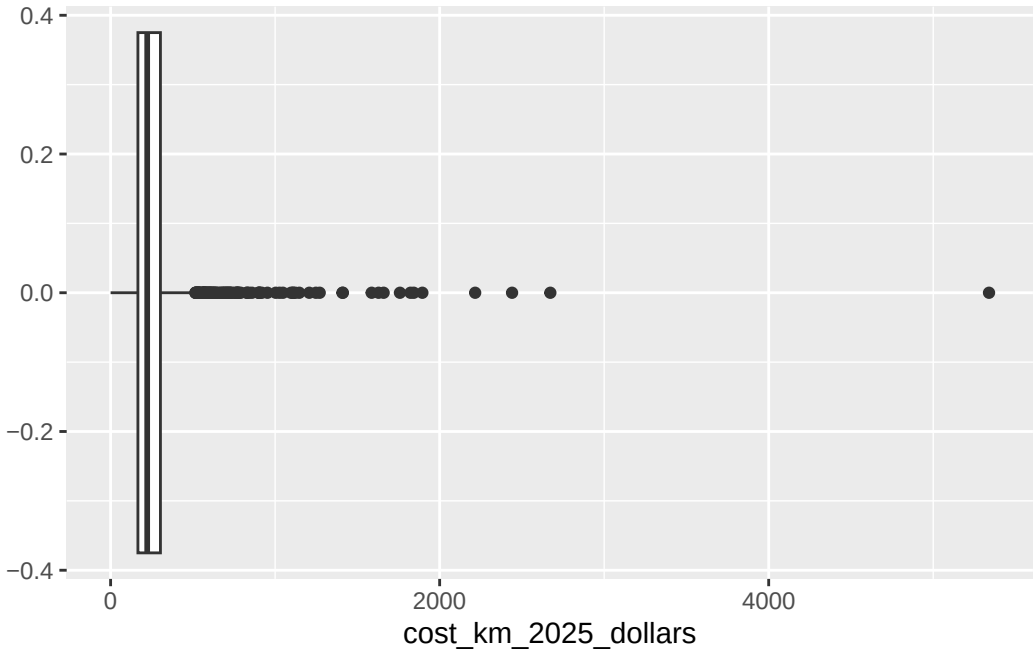

```
# Country distribution
joined_2 %>%
  ggplot(aes(x = "", fill = iso3c)) +
  geom_bar(width = 1) +
  coord_polar(theta = "y") +
  labs(fill = NULL) +
  theme_void()
```



Graveyard

Code Dump

```
ggplot(joined_2_trim, aes(x = cost_km_2025_dollars)) +
  geom_boxplot()
```



Diagnostics/bug-finding code used

```
diesel_b2016_data_1 %>%
  count(iso3c, year) %>%
  filter(n > 1)
```

```
# A tibble: 0 x 3
# i 3 variables: iso3c <chr>, year <dbl>, n <int>
```

```
joined %>% filter(!is.na(diesel_usd_per_liter_a), !is.na(diesel_usd_per_liter_b))
```

```
 [1] id                          iso3c
 [3] city                        line
 [5] phase                       rr
 [7] length                      tunnel
 [9] elevated                    atgrade
[11] stations                    platform_length_meters
[13] year                        cost_km_2025_dollars
[15] legal_systems               mixed_systems_components
[17] col_ruler_iso3c             government_effectiveness
```

```
[19] control_of_corruption           political_stability
[21] regulatory_quality              rule_of_law
[23] voice_and_accountability        gdp_per_capita_constant_2015_us
[25] trade_percent_of_gdp            wage_per_month
[27] diesel_usd_per_liter_a          diesel_usd_per_liter_b
<0 rows> (or 0-length row.names)
```

```
joined_2_trim %>%
  distinct(iso3c, year, diesel_usd_per_liter) %>%
  ggplot(aes(x = year, y = diesel_usd_per_liter, color = iso3c)) +
  geom_line()
```

```
Warning: Removed 50 rows containing missing values or values outside the scale range
(`geom_line()`).
```

